

Email Forensics & Threat Analysis Report

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Email Header Forensics Report

Introduction:

This report presents a forensic investigation of a suspicious email titled "Farewell." The analysis focuses on validating email authenticity, examining transmission headers, tracing the originating infrastructure, and assessing potentially harmful or threatening content embedded within the message.

Objective:

To analyze the authenticity, transmission path, and content of the email in order to identify indicators of compromise (IOCs), detect spoofing attempts, and assess potential threats or malicious intent.

Tools Used:

- MXToolbox
- MHA Header Analyzer
- VirusTotal
- IPTrackerOnline
- Base64 Decoder
- CentralOps

Methodology:

1. Email Header Analysis
2. Relay Path Examination
3. SPF/DKIM/DMARC Validation
4. Originating IP Investigation
5. Payload Decoding
6. Threat Content Assessment

Basic information:

Email from **Boyce Spears** to **Kindi Bullard**, titled "**Farewell**", sent on **April 22, 2025, at 14:34 UTC**.

From: Boyce Spears <ullboycespears@protonmail.com>

To: "Kindi.bullard@yahoo.com" <Kindi.bard@yahoo.com>

Subject: Farewell

Message Id:

<0SSOARYrMcAEQ3rmZDTT9_YuBiJi60TOhXQsRpSXkDrwSW8bmVhavFnys21UhDi0S6YhKz6gHVQJw7k_ZmT1peZ2foZVtjuv6lhJnIetXI4=@protonmail.com>

Creation time: Tue, 22 Apr 2025 14:34:08 +0000

Summary	
Subject	Farewell
Message Id	<0SSOARYrMcAEQ3rmZDTT9_YuBIU60TOhXQsRpSXkDrwSW8bmVhavFnys21UhDi056YhKz6gHVQW7k_ZmT1peZ2foZVtjuv6lhJnletXI4=@protonmail.com>
Creation time	Tue, 22 Apr 2025 14:34:08 +0000
From	Boyce Spears <boycespears@protonmail.com>
To	"Kindi.bullard@yahoo.com" <Kindi.bullard@yahoo.com>

Figure 1: basic info

Executive Summary:

A comprehensive header analysis was conducted on the “Farewell” email. All major authentication mechanisms—SPF, DKIM, and DMARC—are valid and strictly aligned. The relay path is minimal and entirely within ProtonMail’s infrastructure. No signs of header tampering, relay hijacking, or spoofing were found. The message can be reliably traced to the originating client IP 109.224.244.24.

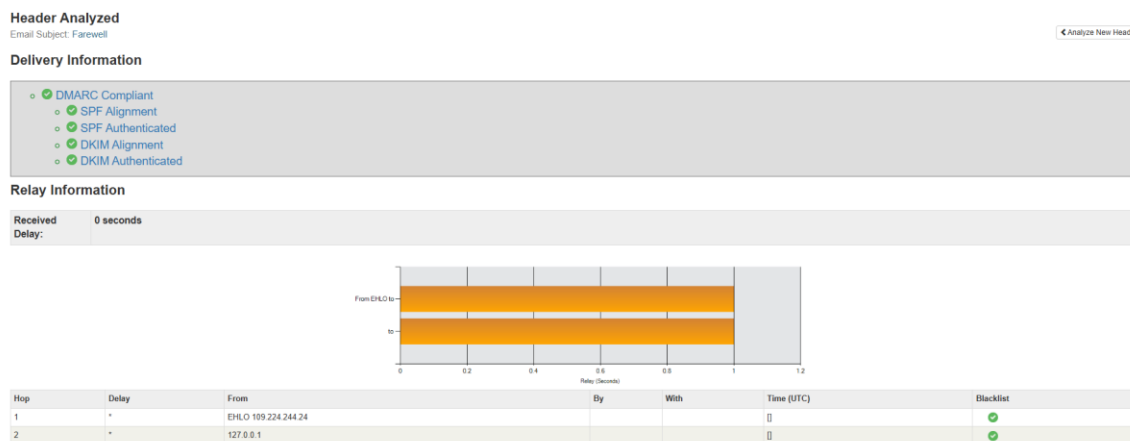


Figure 2: Header Analyzed by mxtoolbox

Detailed Findings:

Relay Path Analysis

Two hops were observed:

Ho p	Client IP / Host	Receiving MTA	Delay	Timestamp (UTC)	Blacklist Status	Type
1	109.224.244.24 (EHLO mail-24424.protonmail.ch)	10.253.62.170	0 s	2025-04-22 14:34:08	Not listed	SMTPs (version=TLS1_3 cipher=TLS_AES_128_GCM_SHA256)

2	127.0.0.1 (loopback)	atlas-production.v2-mail-prod1-gg1.omega.yahoo.com pod-id atlas--production-gg1-8654446fdb-kd76r.gg1.yahoo.com	0 s	2025-04-22 14:34:08	Not listed	HTTP
---	-------------------------	---	-----	------------------------	------------	------

Hop	Submitting host	Receiving host:	Time	Delay	Type
1	109.224.244.24 (EHLO mail-24424.protonmail.ch)	10.253.62.170	4/22/2025 8:34:13 PM		SMTPs (version=TLS1_3 cipher=TLS_AES_128_GCM_SHA256)
2	127.0.0.1	atlas-production.v2-mail-prod1-gg1.omega.yahoo.com pod-id atlas--production-gg1-8654446fdb-kd76r.gg1.yahoo.com	4/22/2025 8:34:13 PM	0 seconds	HTTP

Figure 3: receive header info by mha.azurewebsites.net

Authentication Results:

1.SPF

Passed (aligned with domain protonmail.com)

spf:protonmail.com:109.224.244.24

Hide

Solve Known Delivery Problems

v=spf1 include:_spf.protonmail.ch ~all

Prefix	Type	Value	PrefixDesc	Description
	v	spf1		The SPF record version
+	include	_spf.protonmail.ch	Pass	The specified domain is searched for an 'allow'.
~	all		Soft fail	Always matches. It goes at the end of your record.

Test		Result
✓	SPF Record Published	SPF Record found
✓	SPF Record Deprecated	No deprecated records found
✓	SPF Multiple Records	Less than two records found
✓	SPF Alignment	Domain found in SPF
✓	SPF Contains characters after ALL	No items after 'ALL'
✓	SPF Syntax Check	The record is valid
✓	SPF Included Lookups	Number of included lookups is OK
✓	SPF Recursive Loop	No Recursive Loops on Includes
✓	SPF Duplicate Include	No Duplicate Includes Found
✓	SPF Type PTR Check	No type PTR found
✓	SPF Void Lookups	Number of void lookups is OK
✓	SPF MX Resource Records	Number of MX Resource Records is OK
✓	SPF Record Null Value	No Null DNS Lookups found
✓	SPF Authentication	SPF Passed for IP
✓	DMARC Record Published	DMARC Record found
✓	DMARC Policy Not Enabled	DMARC Quarantine/Reject policy enabled

Reported by mx3.protonmail.com on 5/23/2025 at 3:37:33 AM (UTC 0). Just for you

Trans...

Figure 3.1: spf info by mxtoolbox

Record: v=spf1 include:_spf.protonmail.ch ~all

Outcome: Pass for IP 109.224.244.24

Alignment: Strict (sender domain exactly matches “protonmail.com”).

Conclusion: The IP is explicitly authorized; SPF is correctly implemented.

2.DKIM

Valid (RSA-2048 key, strict canonicalization)

Headers Found	
Header Name	Header Value
Return-Path	<boycespears@protonmail.com>
X-Originating-Ip	[109.224.244.24]
Received-SPF	pass (domain of protonmail.com designates 109.224.244.24 as permitted sender)
Authentication-Results	alias-production v2-mail-prod1-gq1 omega yahoo.com;
X-Apparently-To	kindi.bullard@yahoo.com, Tue, 22 Apr 2025 14:34:13 +0000
X-MailSG	cnbNsgoWLDuk1AO6Fx_9hzWC1479mZVSYn9.x7WauZvmYQ
DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=protonmail.com; s=protonmail3; t=1745332451; x=1745591651; bh=HxL4Y9MhmZqAhpPXaRv1KD1aR4VEPLHLDYdD0mXIA=; h=Date: To From: Subject: Message-ID: Feedback-ID: From: To: Cc: Date: Subject: Reply-To: Feedback-ID: Message-ID: BIMI-Selector: List-Unsubscribe: Post; b=sMjphnDEHFr7Kc4vYiqQTLZRs8pC8K3V2bVoNqJnKc5mTfwbasPvJmXuoRq pWzKLQ9xE7q0ugeRv6RR9nuW3YOLUq/czFRpMR9vieB7ycQs9pY137ENd QXldgJUV e1Uv67VZLcUEKJmKDVLoK8TmPwtgDb+pr9OMuJ7LhUE/Qw527hsaRVKvgbQwlsF cKdYLTbKLSvHlHfRHYLaHfYUD5czX7wqjgR9d45F+6DXRRNo4NdlJlD2z4UfDcS 95zhfbaCakomyISzCrydwf0mftqz8bVMIGsL23ULd8mgXO4mH
Date	Tue, 22 Apr 2025 14:34:08 +0000
To	"Kindi.bullard@yahoo.com" <Kindi.bullard@yahoo.com>
From	Boyce Spears <boycespears@protonmail.com>
Subject	Farewell
Message-ID	<D5SOARYMcAEQ3mZDTT9_YuBU60TChXQaRp5XaDrwSW8tmVhavFrys21UhdK056YhKz9gIvQJw7K_ZmT1peZ2bzVljudfHJnt04=@@protonmail.com>
Feedback-ID	141478856:user:proton
X-Fm-Message-ID	aa151fb1138228352adc9a172ac0fb6c8907b0
MIME-Version	1.0
Content-Type	multipart/alternative;
Content-Length	7568

Figure 4: Header info by mxtoolbox

Field	Value	Forensic Significance
Return-Path	<boycespears@protonmail.com>	Bounce address aligns with sender domain.
X-Originating-IP	[109.224.244.24]	Public client IP; geolocation may yield sender’s region.
Authentication-Results	SPF/DKIM/DMARC pass by Yahoo MX	End-recipient authentication verdict.
X-Apparently-To	kindi.bullard@yahoo.com; 22 Apr 2025 14:34:13 +0000	Confirms delivery to recipient.
Feedback-ID	141478856:user:proton	Internal tracking for abuse correlation.
Content-Type	multipart/alternative; Length=7568	Plaintext & HTML variants present.
From	Boyce Spears <boycespears@protonmail.com>	The visible sender address to the recipient; critical for DMARC alignment.
To	"Kindi.bullard@yahoo.com" <Kindi.bullard@yahoo.com>	The primary recipient of the email.
Subject	Farewell	The subject line of the email.

Probable Originating IP Analysis:

The IP address 109.224.244.24 is owned by Proton AG (ProtonMail), based in Plan-les-Ouates, Geneva, Switzerland. The IP aligns with ProtonMail infrastructure, confirming sender authenticity.

Probable originating IP address: 109.224.244.24			
IP Address	109.224.244.24	City	Plan-les-Ouates
Hostname	Proton AG	Country	Switzerland
ISP	Proton AG	Latitude	46.166672
Region	Geneve	Longitude	6.11667
Country Flag			

Figure 5: IP Analysis by iptrackeronline

G8gc2F5IGdvb2RieWUuClldvXIgbW9tIGhhZCBhIGJveWZyaWVuZCB3aG8gc2hvdCB0aW1zZWxmLgpZb3VyIG1vbSB3YXMGYSBmb3N0ZXIgcGFyZW50IHdoawNoIGJyb3VnaHQga2lkcyBpbiBhbmQgb3V0IG9mIHlvdXIgaG9tZS4KWW91ciBzdGVwIGRhZCBkaWVklGpZb3VyIHN0ZXAgbW9tIGRpZWQuClldvXIgZnJpZW5kIHdhcyBraWxsZWQgaW4gYSBkcnuayBjYXlgaWNjaWRlbnQgd2hlbiB5b3Ugd2VyZSBzdXBwb3NlZCB0byBoYW5nb3V0IGxhdGVyLgpZb3VyIGRhZCBkaWVklGpZb3VyIG1vbSBkaWVklGpZb3VyIGdyYW5kbWEgZGllZC4KWW91ciBuaWVjZSBkaWVklGpZb3VyIGdyYW5kbWEgZGllZC4KWW91ciBjbGFzc21hdGUgZGllZCBpbiBhIG1vdG9yY3ljGUgYWNjaWRlbnQuClldvXIgY29sbGVnZSB0ZWftbWF0ZSBkaWVklGp1bXBpbmcgb3V0IG9mIGEgdmVoaWNsZS4KWW91ciBjaGlsZGhvb2QgZnJpZW5kIGtpbGxlZCB0aW1zZWxmLgpZb3VyIHNpc3RlciB0cmllZCB0byBraWxsIGhlcniBjYgYWdhaW4uClldvXIgdW5jbGUgZGllZC4KWW91ciBoaWdoIHNaG9vbCBjb2FjaCBkaWVklGpZb3VyIGNoaWxkaG9vZCBmcmlbmQgZGllZC4KCKl0IGxvb2tzIGxp2UgZGVhdGgaXMGd2hhdCBmb2xsb3dzIHlvdSBhcm91bmQuIExxc2luZyBwZW9wbGUgb3V0IG9mIHlvdXIgbGlmZSBpcyB3aGF0IHlvdSBhcmUgdXNlZCB0bywgc28gbG9zaW5nIGZyaWVuZHMgb3ZlciBhIGxpHRsZSB0YXJtbGVzcyBmdW4gSSBoYWQgc2hvdWxkbid0IGJlIHrvbyBtdWNoIG9uIHlvdS4gRG8geW91IGtub3cgaG93IGVhc3kgaXQgaXMGdG8gZ2V0IGluZm9ybWFOaW9uIGFib3V0IHBlb3BsZT8gSXQgaXMGZWFzeSB0byBnZXQgaW50byBzb21lb25lJ3MgcGVyc29uYWwgZGF0YSwgYnV0IGV2ZlXJ5dGhpbmcgaXMGb3V0IHRoZXJlIG9uIHRoZSBpbnRlcm5ldCBpZiB5b3UgbG9vay4gSSBjb3VsZCB0YXZlIHRhbGtlZCBtb3JlIGFib3V0IHlvdXIgZmFtaWx5IGlzc3VlcyB5b3UgZGVhbHQgd2l0aCBncm93aW5nIHVwIGxp2UgeW91ciBEYWQncywgYnV0IEkgd29uJ3QuIFlvdSBjYW4gc2VhcmNoIHRoZSBpbnRlcm5ldCBpZiB5b3UgZG9uJ3Qga25vdyB3aGF0IEkgYW0gcmlmVmZXJyaW5nIHRvby4gWW91IHRoaW5rIHlvdSBvdXQgc21hcnRlZCBtZSBieSBzaHV0dGluZyBvZmYgeW91ciBwaG9uZSBudW1iZXlgaW5kIGxvY2tpbmcgYXBwbGUgb3V0LiBjIHdhcyBhbHJlYWR5IGFoZWfkIG9mIHlvdS4gSXQgd2FzIG15IHBSZWfzdXJlgoKSXQgZG9lc24ndCBwYXkgdG8gYmUgYSBnb29kIHBlcnNvb29kSXQgZG9lc24ndCBnZXQgeW91IGFueXdoZXJlLiBzb3UgbmVlZCB0byBnbyB0byBTYW1zdW5nLCBbcHBsZSBsaWVklLiBUaGV5IGFyZSBub3QgdGhlIG1vc3Qgc2VjdXJlLiBUaGUgaW50ZXJuZXQgaXMGbm90IHNIY3VyZS4KCKJveWNl

The screenshot shows the CyberChef web application interface. The 'Recipe' panel on the left has 'From Base64' selected. The 'Input' panel contains a long Base64 string. The 'Output' panel shows the decoded text, which is a poem about a person's life.

Recipe: From Base64

Alphabet: A-Za-z0-9+/=

☒ Remove non-alphabet chars ☐ Strict mode

Input: dXAAd2l0aCBubyBncmFuZHBhcwpCb3RoIHlvdXIgbW9tJ3Mgc2lzdGVycyBkaWVklGJlZm9yZSB5b3UgaG10IghpZ2ggc2Nob29sc1lvdSBoYWQgYSB0ZWftbWF0ZSBkaWUgaW4gYSBjYXlgaWNjaWRlbnQgd2hlbiB5b3Ugd2VyZSBzdXBwb3NlZCB0byBoYW5nb3V0IGxhdGVyLgpZb3VyIGRhZCBkaWVklGpZb3VyIG1vbSB3YXMGYSBmb3N0ZXIgcGFyZW50IHdoawNoIGJyb3VnaHQga2lkcyBpbiBhbmQgb3V0IG9mIHlvdXIgaG9tZS4KWW91ciBzdGVwIGRhZCBkaWVklGpZb3VyIHN0ZXAgbW9tIGRpZWQuClldvXIgZnJpZW5kIHdhcyBraWxsZWQgaW4gYSBkcnuayBjYXlgaWNjaWRlbnQgd2hlbiB5b3Ugd2VyZSBzdXBwb3NlZCB0byBoYW5nb3V0IGxhdGVyLgpZb3VyIGRhZCBkaWVklGpZb3VyIG1vbSBkaWVklGpZb3VyIGdyYW5kbWEgZGllZC4KWW91ciBjbGFzc21hdGUgZGllZCBpbiBhIG1vdG9yY3ljGUgYWNjaWRlbnQuClldvXIgaG9tZS4KWW91ciBjaGlsZGhvb2QgZnJpZW5kIGtpbGxlZCB0aW1zZWxmLgpZb3VyIHNpc3RlciB0cmllZCB0byBraWxsIGhlcniBjYgYWdhaW4uClldvXIgdW5jbGUgZGllZC4KWW91ciBoaWdoIHNaG9vbCBjb2FjaCBkaWVklGpZb3VyIGNoaWxkaG9vZCBmcmlbmQgZGllZC4KCKl0IGxvb2tzIGxp2UgZGVhdGgaXMGd2hhdCBmb2xsb3dzIHlvdSBhcm91bmQuIExxc2luZyBwZW9wbGUgb3V0IG9mIHlvdXIgbGlmZSBpcyB3aGF0IHlvdSBhcmUgdXNlZCB0bywgc28gbG9zaW5nIGZyaWVuZHMgb3ZlciBhIGxpHRsZSB0YXJtbGVzcyBmdW4gSSBoYWQgc2hvdWxkbid0IGJlIHrvbyBtdWNoIG9uIHlvdS4gRG8geW91IGtub3cgaG93IGVhc3kgaXQgaXMGdG8gZ2V0IGluZm9ybWFOaW9uIGFib3V0IHBlb3BsZT8gSXQgaXMGZWFzeSB0byBnZXQgaW50byBzb21lb25lJ3MgcGVyc29uYWwgZGF0YSwgYnV0IGV2ZlXJ5dGhpbmcgaXMGb3V0IHRoZXJlIG9uIHRoZSBpbnRlcm5ldCBpZiB5b3UgbG9vay4gSSBjb3VsZCB0YXZlIHRhbGtlZCBtb3JlIGFib3V0IHlvdXIgZmFtaWx5IGlzc3VlcyB5b3UgZGVhbHQgd2l0aCBncm93aW5nIHVwIGxp2UgeW91ciBEYWQncywgYnV0IEkgd29uJ3QuIFlvdSBjYW4gc2VhcmNoIHRoZSBpbnRlcm5ldCBpZiB5b3UgZG9uJ3Qga25vdyB3aGF0IEkgYW0gcmlmVmZXJyaW5nIHRvby4gWW91IHRoaW5rIHlvdSBvdXQgc21hcnRlZCBtZSBieSBzaHV0dGluZyBvZmYgeW91ciBwaG9uZSBudW1iZXlgaW5kIGxvY2tpbmcgYXBwbGUgb3V0LiBjIHdhcyBhbHJlYWR5IGFoZWfkIG9mIHlvdS4gSXQgd2FzIG15IHBSZWfzdXJlgoKSXQgZG9lc24ndCBwYXkgdG8gYmUgYSBnb29kIHBlcnNvb29kSXQgZG9lc24ndCBnZXQgeW91IGFueXdoZXJlLiBzb3UgbmVlZCB0byBnbyB0byBTYW1zdW5nLCBbcHBsZSBsaWVklLiBUaGV5IGFyZSBub3QgdGhlIG1vc3Qgc2VjdXJlLiBUaGUgaW50ZXJuZXQgaXMGbm90IHNIY3VyZS4KCKJveWNl

Output: Here is some last dirt I have for you.

Death:

You grew up with no grarldpas
Both your mom's sisters died before you hit high school
You had a teammate die in a car crash
Your best friend and your first girlfriend almost got killed by her mom. This lead her to getting moved to a by DCFs with no chance to say goodbye.
Your mom had a boyfriend who shot himself.
Your mom was a foster parent which brought kids in and out of your home.
Your step dad died.

Figure 7: decode base64 strings

Decoded Payload Summary:

Here is some last dirt I have for you.

Death:

You grew up with no grandpas

Both your mom's sisters died before you hit high school

You had a teammate die in a car crash

Your best friend and your first girlfriend almost got killed by her mom. This lead her to getting moved to a by DCFs with no chance to say goodbye.

Your mom had a boyfriend who shot himself.

Your mom was a foster parent which brought kids in and out of your home.

Your step dad died.

Your step mom died.

Your friend was killed in a drunk car accident when you were supposed to hangout later.

Your dad died.

Your mom died.

Your grandma died.

Your niece died.

Your grandma died.

Your classmate died in a motorcycle accident.

Your college teammate died jumping out of a vehicle.

Your childhood friend killed himself.

Your sister tried to kill herself again.

Your uncle died.

Your high school coach died.

Your childhood friend died.

It doesn't pay to be a good person. It doesn't get you anywhere. You need to go to Samsung, Apple lied. They are not the most secure. The internet is not secure.

Encoding strings:

[illegible]

Decoded Payload Summary:

Here is some last dirt I have for you. Death: You grew up with no grandpas Both your mom's sisters died before you hit high school You had a teammate die in a car crash Your best friend and your first girlfriend almost got killed by her mom. This lead her to getting moved to a by DCFs with no chance to say goodbye. Your mom had a boyfriend who shot himself. Your mom was a foster parent which brought kids in and out of your home. Your step dad died. Your step mom died. Your friend was killed in a drunk car accident when you were supposed to hangout later. Your dad died. Your mom died. Your grandma died. Your niece died. Your grandma died. Your classmate died in a motorcycle accident. Your college teammate died jumping out of a vehicle. Your childhood friend killed himself. Your sister tried to kill herself again. Your uncle died. Your high school coach died. Your childhood friend died. It looks like death is what follows you around. Losing people out of your life is what you are used to, so losing friends over a little harmless fun I had shouldn't be too much on you. Do you know how easy it is to get information about people? It is easy to get into someone's personal data, but everything is out there on the internet if you look. I could have talked more about your family issues you dealt with growing up like your Dad's, but I won't. You can search the internet if you don't know what I am referring too. You think you out smarted me by shutting off your phone number and locking apple out. I was already ahead of you. It was my pleasure. It doesn't pay to be a good person. It doesn't get you anywhere. You need to go to Samsung, Apple lied. They are not the most secure. The internet is not secure. Boyce

Decoded Content Analysis:

The base64-decoded payload contained psychologically manipulative and threatening language directed toward the recipient. The message referenced personal tragedies, implied unauthorized access to personal information, and attempted emotional intimidation.

The content demonstrates indicators consistent with harassment, cyberstalking, and social engineering behavior.

Indicators of Compromise (IOCs)

- Originating IP: 109.224.244.24
- Email Provider: ProtonMail
- Sender Domain: protonmail.com
- Authentication Status: SPF/DKIM/DMARC passed
- Suspicious Indicators:
 - Psychological intimidation
 - Harassment-related language
 - Potential social engineering behavior

Conclusion

The email in question is technically authentic, with all header-based authentication mechanisms confirming its legitimacy. The message originated within ProtonMail's infrastructure, and no technical evidence of spoofing or relay manipulation exists.

However, the decoded message content is disturbing and potentially harmful, containing targeted emotional abuse and intimidation. This elevates the incident from a technical investigation to a potentially criminal matter. It is strongly recommended that legal and mental health professionals be engaged to support the recipient and address any ongoing threats.

Prepared By:

M. Usamanur Rahman Biplob

Role: Digital Forensic Investigator

Investigation Date: May 2025